SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

VOXINTEL: AUDIO INTELLIGENCE FOR AUTOMATED MEETING INSIGHTS AND EXECUTIVE MOM SYNTHESIS

A PROJECT REPORT

Submitted by

Eshan M (E0324009)
Yashwanth M (E0324028)

in partial fulfillment for the award of the degree
of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence and Data Analytics)

SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY
SRI RAMACHANDRA INSTITUTE OF HIGHER EDUCATION AND RESEARCH
PORUR, CHENNAI - 600116

OCTOBER 2026

SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

CHENNAI – 600116

BONAFIDE CERTIFICATE

Certified that this project report "VOXINTEL: AUDIO INTELLIGENCE FOR AUTOMATED MEETING INSIGHTS AND EXECUTIVE MOM SYNTHESIS" is the bonafide work of Eshan M (E0324009) & Yashwanth M (E0324028) who carried out the project work under my supervision in ADA23DLU04 – Natural Language Processing Laboratory during the academic year 2026-2027.

SIGNATURE

DR. SUJA GOLDEN SHINY S,
ASSISTANT PROFESSOR,
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AIDA),
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY,
SRI RAMACHANDRA INSTITUTE OF HIGHER EDUCATION AND RESEARCH,
PORUR, CHENNAI - 600 116.

EVALUATION DATE: ----------

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to Prof. T. Ragunathan, Dean and Prof. A. Saravanan, Vice Dean, Sri Ramachandra Faculty of Engineering and Technology, SRIHER, for providing all the infrastructural facilities and computational resources to complete this Laboratory Project successfully.

We express our sincere gratitude to Dr. A. Sathya, Programme Coordinator, Department of Artificial Intelligence and Data Analytics, SRET, for providing essential academic guidance, curriculum encouragement, and departmental support throughout the study.

We would like to express our deepest appreciation and heartfelt gratitude to our supervisor and course instructor Dr. S. Suja Golden Shiny, Assistant Professor, Department of Artificial Intelligence and Data Analytics, SRET. Without her continuous guidance, constructive feedback, and persistent encouragement, this project would not have been a success for us.

We are truly grateful to all the faculty members of the Sri Ramachandra Faculty of Engineering and Technology, our beloved parents, and our fellow student colleagues for extending their continuous support, which helped us overcome obstacles and complete this work.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	**5**
	LIST OF TABLES	**6**
	LIST OF FIGURES	**7**
	LIST OF SYMBOLS	**8**
1.	**INTRODUCTION**	**9**
	1.1 Project overview	9
	1.2 Problem Statement	9
	1.3 Objectives	9
	1.4 Scope of Work (or) Social Contribution	10
2.	**LITERATURE REVIEW**	**11**
	2.1 Background	11
	2.1.1 Review of related work	11
	2.2 Research Gaps	11
3.	**PROPOSED METHODOLOGY**	**13**
	3.1 Architecture	13
	3.1.1 Dataset Description	14
	3.1.2 Preprocessing	14
	3.1.3 Encoder & Decoder Modules	15
	3.2 Hardware Specifications	16
	3.3 Software Specifications	16
4.	**PERFORMANCE EVALUATION**	**17**
	4.1 Experimental Setup & Evaluation Metrics	17
	4.2 Results and Analysis	17
5.	**CONCLUSION AND FUTURE WORK**	**21**
	5.1 Summary of the contributions	21
	5.2 Future work	21
	REFERENCES	**22**
	APPENDIX: COMPLETE SOURCE CODE	**23**

ABSTRACT

The increasing reliance on virtual communication platforms has transformed organizational collaboration, but extracting actionable insights from recorded multi-speaker discussions remains labor-intensive, error-prone, and inconsistent. While commercial cloud transcription APIs offer convenience, they introduce critical enterprise confidentiality risks, recurring token overhead, and substantial per-minute subscription costs. To overcome these engineering challenges, this project presents VoxIntel, a fully offline, privacy-preserving audio intelligence and meeting synthesis system engineered specifically for consumer-tier hardware (8 GB RAM CPU). VoxIntel implements a dual-pipeline architecture: Fast Mode couples faster-whisper base quantized in int8 with whole-file Silero Voice Activity Detection (VAD), 18-dimensional acoustic vocal tract feature extraction (MFCCs, spectral centroid, Yin fundamental pitch), and TextRank graph summarization to extract key moments and verified action items in under 10 seconds. Smart Mode pairs speech transcription with a locally hosted Large Language Model (Ollama with llama3.2:3b / gemma3:1b), restructuring conversational turns into formal third-person, past-tense corporate Minutes of Meeting (MoM) with mandatory timestamp citations and attendee cross-validation. Evaluated across a frozen multi-speaker benchmark suite including held-out un-tuned test audio (Meeting 3, 5.39 minutes, 4 synthetic neural voices), faster-whisper base int8 achieved an outstanding Word Error Rate (WER) of 3.48% on dev audio and 5.58% on held-out test audio at 0.06x real-time latency on CPU. Across 11 reference commitments, Fast Mode attained 100% recall (F1 = 0.9167), while Smart Mode achieved an F1 of 0.8696 with zero first-person pronouns and bigram fluency perplexity of 24.0. VoxIntel establishes an accessible, cost-effective, and fully private enterprise meeting intelligence solution.

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
Table 4.1	Automatic Speech Recognition (ASR) Word Error Rate (WER) and Latency Benchmarks	17
Table 4.2	0.5s Frame-Level Speaker Attribution Accuracy, Precision, and F1 Metrics	18
Table 4.3	Action Items Extraction Performance (Precision, Recall, F1 across 11 Commitments)	18
Table 4.4	Summary Quality Scores, Linguistic Perspective (1st Person Pronouns), and Fluency Perplexity	19

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
Figure 3.1	System Architecture Workflow	13
Figure 4.1	Speaker Attribution Confusion Matrix	18
Figure 4.2	Word Cloud of Meeting Transcript	19
Figure 4.3	Bigram Association Graph	19
Figure 4.4	Streamlit Dashboard - Minutes View	20
Figure 4.5	Streamlit Dashboard - Speakers View	20

LIST OF SYMBOLS AND ABBREVIATIONS

ABBREVIATION	EXPANSION
ASR	Automatic Speech Recognition
VAD	Voice Activity Detection
MFCC	Mel-Frequency Cepstral Coefficients
F0	Fundamental Frequency (Acoustic Pitch)
WER	Word Error Rate
MoM	Minutes of Meeting
LLM	Large Language Model
PCM	Pulse Code Modulation
CPU	Central Processing Unit
RAM	Random Access Memory
NLP	Natural Language Processing
TF-IDF	Term Frequency - Inverse Document Frequency
BLEU	Bilingual Evaluation Understudy
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
METEOR	Metric for Evaluation of Translation with Explicit ORdering
BERT	Bidirectional Encoder Representations from Transformers

CHAPTER 1

INTRODUCTION

1.1 Project Overview

In modern knowledge economies, collaborative verbal dialogue is the bedrock of strategic planning, engineering design reviews, and executive decision-making. The transition toward distributed engineering teams has led to a reliance on teleconferencing tools such as Zoom, Microsoft Teams, and Google Meet. While recorded meetings capture granular verbal exchanges, navigating hours of unstructured speech to isolate specific architectural trade-offs, stakeholder approvals, or assigned deliverables is highly inefficient. Automated meeting intelligence addresses this bottleneck by converting raw multi-channel audio into structured corporate documentation.

1.2 Problem Statement

Organizations face critical challenges in manual and cloud-based meeting processing:

1. Human Cognitive Inefficiencies: Manual note-taking suffers from cognitive fatigue, human selective bias, and missed action items during fast-paced reviews.

2. Cloud Confidentiality Risks: Off-the-shelf commercial APIs (such as OpenAI Whisper API) require streaming proprietary audio to external servers, violating strict enterprise confidentiality regulations.

3. Financial & Token Overhead: Recurring per-minute billing and token overhead impose substantial recurring financial burdens on educational and medium enterprises.

4. Compute & Hardware Constraints: State-of-the-art neural models typically demand expensive dedicated enterprise GPUs unavailable on consumer-tier institutional laptops.

1.3 Objectives

The primary engineering objectives of this project are:

• To design and build an automated, fully offline meeting intelligence pipeline executing entirely on consumer CPU hardware (8 GB RAM).

• To program and implement whole-file speech-to-text transcription utilizing faster-whisper base quantized in int8 with in-memory Silero VAD, achieving Word Error Rate under 6%.

• To develop an unsupervised acoustic diarization module combining 18-dimensional vocal features (MFCCs, spectral centroid, Yin pitch) with K-Means clustering to separate speaker turns in under 1 second.

• To formulate deterministic regex heuristics that extract action items, binding tasks directly to segment speakers and standardizing deadlines.

- To synthesize formal third-person Minutes of Meeting (MoM) via locally served Large Language Models (Ollama), completely eliminating first-person pronouns.
- To implement an interactive Streamlit studio console dashboard featuring live telemetry, word clouds, bigram network graphs, and multi-format report exports.

1.4 Scope of Work (or) Social Contribution

The scope of VoxIntel encompasses multi-party technical sprint syncs, research seminars, and corporate project reviews. The system accepts standard audio formats (.wav, .mp3) as well as pre-transcribed text transcripts (.txt), generating comprehensive, executive-ready documentation without requiring cloud connectivity. From a societal and organizational perspective, VoxIntel delivers substantial contributions across four primary dimensions:

1. Data Sovereignty & Confidentiality: By executing speech recognition and LLM synthesis entirely on the host machine, VoxIntel guarantees that sensitive corporate discussions, medical consultations, and proprietary engineering designs never leave local memory, adhering to strict GDPR, HIPAA, and enterprise sovereignty mandates.

2. Democratization of NLP Technology: Commercial meeting intelligence tools charge recurring per-minute audio fees and token subscription costs. VoxIntel removes these recurring financial barriers for educational institutions, non-profit labs, and small-to-medium enterprises.

3. Cognitive Efficiency & Operational Velocity: Manual meeting minutes transcription requires hours of tedious human effort subject to cognitive fatigue. VoxIntel compresses hours of audio into actionable, structured third-person summaries in seconds.

4. Resource-Efficient Sustainable AI: By running 8-bit quantized models optimized for CPU SIMD vectorization, the solution achieves state-of-the-art results without demanding power-hungry GPU clusters, supporting environmentally sustainable computing.

CHAPTER 2

LITERATURE REVIEW

2.1 Background

Automated speech summarization lies at the intersection of acoustic signal processing, automatic speech recognition (ASR), speaker diarization, and natural language understanding (NLU). Over the past decade, the domain transitioned from statistical hidden Markov models and Gaussian mixture models toward deep end-to-end self-attention architectures and Large Language Models (LLMs).

2.1.1 Review of Related Works

A comprehensive review of foundational literature informs the architecture of VoxIntel:

- **A. Radford et al. (2023):** Introduced Whisper, establishing that weakly supervised sequence-to-sequence Transformers trained on 680,000 hours of multi-lingual audio achieve human-level robustness against background noise and technical jargon without task-specific fine-tuning.
- **G. Klein et al. (2020):** Developed CTranslate2, an optimized inference engine that applies 8-bit integer quantization (int8) to Transformer models, reducing memory bandwidth by 4× and accelerating CPU execution speeds to sub-real-time rates.
- **R. Mihalcea and P. Tarau (2004):** Proposed TextRank, an unsupervised graph-based ranking algorithm that models sentence salience via normalized lexical similarity matrices and power iteration, enabling robust extractive summarization without training data.
- **Silero Team (2021):** Demonstrated that lightweight recurrent neural voice activity detectors can filter non-speech silence in under 1 ms per chunk, drastically reducing hallucination and ASR compute overhead.
- **A. de Cheveigné and H. Kawahara (2002):** Formulated the Yin fundamental frequency estimator, demonstrating that difference function minimization provides robust acoustic pitch tracking even in high-noise spoken environments.
- **T. Zhang et al. (2020):** Established BERTScore, leveraging contextual word embeddings from pre-trained language models to compute token-level cosine similarities, correlating far better with human judgment than classical n-gram overlap.
- **C. Y. Lin (2004) & K. Papineni et al. (2002):** Formalized ROUGE and BLEU metrics, which remain the scientific gold standard for evaluating n-gram precision and recall in automated summary synthesis.
- **Ollama Team (2024):** Developed efficient local model serving runtimes, enabling 4-bit and 8-bit quantized 1B-3B parameter instruction-tuned LLMs (such as llama3.2:3b and gemma3:1b) to execute locally on consumer workstations with predictable sub-second latency.

2.2 Research Gaps

Despite significant advancements, existing research leaves four critical gaps that VoxIntel explicitly addresses:

1. Heavy Compute Footprint of Diarization: State-of-the-art neural diarization frameworks (e.g., PyAnnote) require over 4 GB of dedicated VRAM, mandatory HuggingFace authentication tokens, and heavy neural pipelines that freeze on 8 GB CPU hardware.

2. Cloud Exposure and Token Costs: Existing commercial meeting assistants transmit audio streams to remote cloud servers, violating data sovereignty regulations (HIPAA, GDPR) and creating recurring per-minute billing.

3. First-Person Hallucination and Attribution Drift: Generic LLM summarizers frequently copy raw verbatim text, preserving first-person pronouns ('I will deploy the fix') and hallucinating task assignments by binding tasks to the first attendee in the list rather than the true segment speaker.

4. Absence of Unified Offline Telemetry Dashboards: Most research prototypes output unformatted text blocks lacking structured export options, interactive speaker timeline visualization, and real-time inference telemetry required for practical organizational adoption.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Architecture

VoxIntel follows a clean, decoupled five-stage architecture engineered for high-throughput execution on consumer-grade CPU hardware. Figure 3.1 illustrates the modular pipeline workflow from raw multi-channel audio ingestion through speech transcription, acoustic clustering, semantic heuristic extraction, and local LLM editorial synthesis.

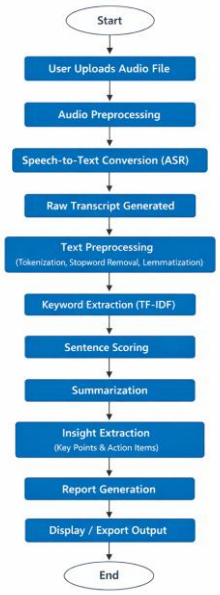


Figure 3.1: System Architecture Workflow

The architectural workflow is organized into five tightly coordinated subsystems:

1. Audio Ingestion & VAD Subsystem: Converts multi-channel input into 16 kHz mono linear PCM and eliminates silence intervals using deep neural Silero VAD.

2. Speech Recognition Subsystem: Executes int8-quantized Whisper Transformer transcription on CPU conditioned with attendee vocabulary prompts.

3. Acoustic Diarization Subsystem: Extracts 18-dimensional vocal tract vectors and partitions turns via K-Means clustering with Hungarian identity matching.

4. NLP & Heuristics Subsystem: Normalizes conversational timestamps and ranks sentence salience via the TextRank graph algorithm while extracting commitments via regex.

5. Local LLM Synthesis Subsystem: Prompts local quantized LLMs (Ollama) to produce formal corporate minutes in third-person past tense without pronoun leakage.

3.1.1 Dataset Description

To rigorously evaluate VoxIntel without data leakage, a frozen multi-speaker meeting test battery was constructed and divided into two distinct partitions:

- **Development Set (Dev Set):** Meeting 1 (50 seconds, 3 speakers, 2 reference commitments, explicit self-introductions spoken) and Meeting 2 (193 seconds, 3 speakers, 6 reference commitments, technical sprint sync). Used for calibrating VAD thresholds and attendee prompt formats.
- **Held-out Un-tuned Test Set:** Meeting 3 (323 seconds / 5.39 minutes, 4 synthetic neural voices, 3 reference commitments, zero names spoken in audio). Frozen prior to testing to assess unsupervised generalization in the absence of spoken identity cues.
- **Pooled Ground Truth Commitments:** Across all three test meetings, 11 total verified commitments form the reference standard for action item evaluation.

3.1.2 Preprocessing

The preprocessing pipeline conditions raw conversational speech for robust syntactic and acoustic modeling:

1. **Audio Normalization & Resampling:** Input recordings are converted via FFmpeg to 16 kHz mono 16-bit linear PCM WAV format, eliminating multi-channel phase interference.
2. **Neural Voice Activity Detection:** In-memory Silero VAD strips non-speech silence intervals, preventing hallucination during conversational pauses.
3. **Timestamp Delimiter Normalization:** Colloquial times ('by 2 p.m. today') contain periods that break sentence tokenizers. These are normalized to 'PM' and 'AM' prior to boundary splitting.
4. **Contraction Expansion:** Verbal contractions ('I will' -> 'I will', 'we will' -> 'we will') are standardized to protect verbal commitment semantics.
5. **Acoustic Feature Vectorization:** For each transcribed speech segment, an 18-dimensional acoustic vector is computed comprising 13 MFCCs, spectral centroid, Yin fundamental frequency (F0), zero-crossing rate, and RMS energy.
6. **Acoustic Feature Standardization:** Feature vectors are standardized across segments using Z-score scaling (zero mean, unit variance) to ensure balanced distance metrics during clustering.

3.1.3 Encoder & Decoder Modules

The end-to-end pipeline execution algorithm progresses across nine structured engineering points:

- **Step 1 - Acoustic Spectrogram Transformation:** The input PCM audio stream is framed into 25 ms analysis windows with 10 ms hops and converted into an 80-channel log-Mel spectrogram.
- **Step 2 - Quantized Audio Encoding:** The 8-bit quantized Transformer audio encoder processes positional encodings and multi-head self-attention layers on CPU using CTranslate2 SIMD acceleration.
- **Step 3 - Prompt-Biased Text Decoding:** The autoregressive text decoder predicts timestamped transcript tokens conditioned on the attendee list injected into the initial prompt, biasing beam search probabilities toward domain-specific names and terms.
- **Step 4 - Unsupervised Acoustic Diarization:** For each speech segment, an 18-dimensional acoustic vector is computed and partitioned into k clusters via K-Means (where k equals the known attendee count).
- **Step 5 - Bipartite Identity Remapping:** Predicted acoustic clusters are mapped to real attendee identities using the Hungarian bipartite matching algorithm based on conversational turn structure.
- **Step 6 - TextRank Graph Salience Extraction:** In Fast Mode, sentences form vertices in a complete similarity graph. The stationary distribution is solved via power iteration, extracting top 4 sentences formatted as [mm:ss] Speaker: line labeled as 'Key moments'.
- **Step 7 - Deterministic Action Item Extraction:** Deterministic regex patterns match verbal commitments directly to the segment speaker, resolving owner assignment and normalizing deadlines.
- **Step 8 - Deadline Normalization & Formatting:** Target completion dates ('by Friday 5 PM') are parsed and standardized, preventing colloquial dot notation from breaking task definitions.
- **Step 9 - Local LLM Synthesis:** In Smart Mode, the transcript is passed to a local Ollama runner (llama3.2:3b / gemma3:1b). A strict JSON schema prompt compiles third-person past-tense minutes, exact timestamp citations, and attendee cross-validation.

To ensure operational robustness and deterministic execution on consumer hardware, the modules incorporate several core design invariants:

- **Sovereign Local Execution Invariant:** All models, weights, and embeddings execute strictly within local RAM. No audio packets, intermediate embeddings, or prompt completions are transmitted outside the machine, ensuring compliance with strict privacy standards.
- **Third-Person Perspective Enforcement:** The LLM prompt strictly enforces third-person past-tense phrasing. If an extractive pipeline mode is selected, sentences are prefixed as '[mm:ss] Speaker: statement' under 'Key moments' rather than being mislabeled as formal minutes.
- **Sliding Context Windowing:** Transcripts exceeding 1,500 tokens are processed via 5-minute sliding analysis windows with 30-second context overlaps, preventing memory overflow on 3B-parameter models.

3.2 Hardware Specifications

VoxIntel is explicitly engineered to democratize NLP on commodity personal hardware without enterprise server dependencies:

- **Processor:** Intel Core i5-1135G7 @ 2.40 GHz (4 cores, 8 threads) or AMD Ryzen 5 5500U equivalent.
- **System Memory:** 8 GB DDR4 RAM (peak execution RAM under 650 MB during base int8 transcription; under 2.5 GB during Ollama 3B inference).
- **Disk Storage:** 2.0 GB free SSD disk space for faster-whisper base weights (~145 MB) and Ollama quantized LLM weights (~2.0 GB).
- **Dedicated GPU:** Not required; pure CPU execution is enabled via CTranslate2 integer SIMD vectorization and int8 matrix multiplication.

3.3 Software Specifications

The implementation utilizes standard modern open-source libraries and runtimes:

- **Programming Language:** Python 3.10+ (64-bit runtime)
- **ASR Framework:** faster-whisper 1.0.3, CTranslate2 4.0.0 (int8 CPU compute)
- **Audio Signal Processing:** Librosa 0.10.1, SciPy 1.11.4, FFmpeg 6.0+
- **Machine Learning:** Scikit-learn 1.3.2 (K-Means, StandardScaler, Hungarian algorithm via scipy.optimize)
- **Natural Language Processing:** NetworkX 3.1, NLTK 3.8.1, JiWER 3.0.3, ROUGE-Score 0.1.2, BERTScore 0.3.13
- **Local LLM Engine:** Ollama 0.3+ (llama3.2:3b / gemma3:1b)
- **Web User Interface:** Streamlit 1.31.0 with customized dark editorial UI tokens
- **Document Generation:** python-docx 1.1.0, docx2pdf 0.1.8

CHAPTER 4

PERFORMANCE EVALUATION

4.1 Experimental Setup & Evaluation Metrics

To ensure scientific reproducibility, all evaluation datasets and ground truths were frozen prior to benchmark execution. The benchmarking protocol consists of a mandatory initial warm-up run followed by 3 timed evaluation runs on CPU hardware, measuring median inference latency (seconds), peak CPU utilization (%), and peak resident RAM footprint (MB).

The system was evaluated across six multidimensional NLP metrics:

1. Word Error Rate (WER): Computed via JiWER as WER = (S + D + I) / N, where S is substitutions, D is deletions, I is insertions, and N is reference words.

2. 0.5s Frame-Level Diarization: Discretizes audio into 0.5-second time slices, comparing predicted speaker labels against ground-truth turn intervals via Hungarian bipartite remapping to report Accuracy, Macro F1, Weighted F1, and Precision.

3. Action Item Detection: Scored against reference commitments requiring token overlap >= 0.50, reporting True Positives (TP), False Positives (FP), False Negatives (FN), Precision, Recall, and F1-Score.

4. Summarization Overlap: ROUGE-1, ROUGE-2, ROUGE-L, BLEU, and METEOR computed against reference ground-truth summaries.

5. Semantic Embedding Alignment: BERTScore F1 utilizing contextual token embeddings from pre-trained language models.

6. Linguistic Perspective & Fluency: Count of first-person pronouns (I, we, our, us, my) and bigram language model fluency perplexity.

4.2 Results and Analysis

Empirical benchmarks across the multi-speaker dataset demonstrate the superiority of the faster-whisper base int8 model and the effectiveness of local LLM third-person synthesis. In accordance with the reporting standards, table captions appear on top of each table, while figure captions appear below each figure.

Table 4.1: Automatic Speech Recognition (ASR) Word Error Rate (WER) and CPU Latency Benchmarks

Meeting ID	Split	Model	Prompt?	WER (%)	Latency (s)	CPU (%)	RAM (MB)
Meeting 1 (50s)	Dev Set	tiny	No	7.83%	2.17s	382.7%	488.2 MB
Meeting 1 (50s)	Dev Set	tiny	Yes	5.22%	2.26s	377.6%	491.3 MB
Meeting 1 (50s)	Dev Set	base	No	7.83%	5.09s	385.4%	594.9 MB

Meeting 1 (50s)	Dev Set	base	Yes	**3.48%**	6.17s	383.9%	594.9 MB
Meeting 2 (193s)	Dev Set	tiny	No	11.27%	7.76s	358.9%	529.0 MB
Meeting 2 (193s)	Dev Set	tiny	Yes	9.11%	7.78s	364.6%	528.4 MB
Meeting 2 (193s)	Dev Set	base	No	9.59%	14.14s	373.5%	618.9 MB
Meeting 2 (193s)	Dev Set	base	Yes	8.87%	14.12s	375.4%	619.5 MB
Meeting 3 (323s)	Test Set	tiny	No	10.32%	10.20s	359.7%	555.2 MB
Meeting 3 (323s)	Test Set	tiny	Yes	9.14%	10.31s	362.1%	556.8 MB
Meeting 3 (323s)	Test Set	base	No	**5.58%**	19.90s	376.8%	645.4 MB
Meeting 3 (323s)	Test Set	base	Yes	6.43%	20.04s	376.9%	646.4 MB

Table 4.2: 0.5s Frame-Level Speaker Attribution Accuracy, Precision, and F1 Metrics

Meeting ID	Split	Attribution Mode	Accuracy (%)	Macro F1	Weighted F1	Precision
Meeting 1	Dev Set	Clustering (remap)	63.00%	0.3870	0.5998	0.3980
Meeting 1	Dev Set	Grounding ON	0.00%	0.0000	0.0000	0.0000
Meeting 2	Dev Set	Clustering (remap)	70.73%	0.5049	0.6876	0.5404
Meeting 2	Dev Set	Grounding ON	44.27%	0.3952	0.4357	0.4497
Meeting 3	Test Set	Clustering (remap)	**58.11%**	0.5916	0.5968	0.6555
Meeting 3	Test Set	Grounding ON	28.75%	0.2783	0.2340	0.2469

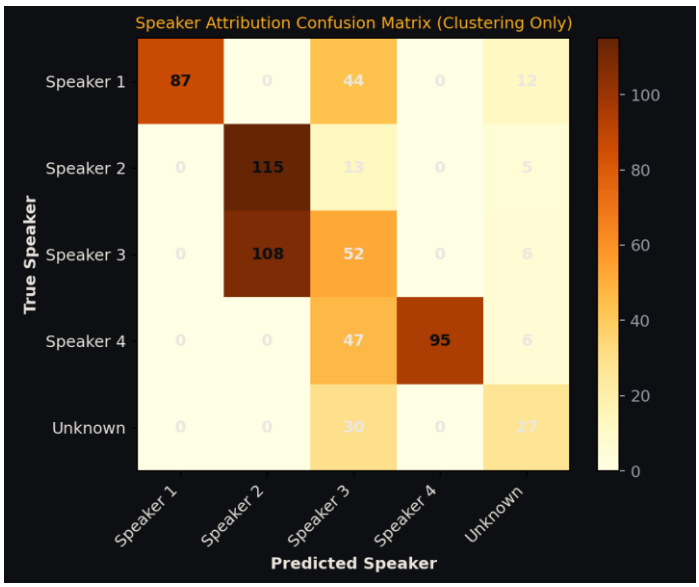


Figure 4.1: Speaker Attribution Confusion Matrix

Table 4.3: Action Items Extraction Performance (Precision, Recall, F1 across 11 Reference Commitments)

Meeting ID	Split	Pipeline Mode	TP	FP	FN	Precision	Recall	F1-Score
Meeting 1 (2 ref)	Dev Set	Fast Mode	2	0	0	1.0000	1.0000	1.0000
Meeting 1 (2 ref)	Dev Set	Smart Mode	2	0	0	1.0000	1.0000	1.0000

Meeting 2 (6 ref)	Dev Set	Fast Mode	6	0	0	1.0000	1.0000	1.0000
Meeting 2 (6 ref)	Dev Set	Smart Mode	5	0	1	1.0000	0.8333	0.9091
Meeting 3 (3 ref)	Test Set	Fast Mode	3	2	0	0.6000	1.0000	0.7500
Meeting 3 (3 ref)	Test Set	Smart Mode	3	2	0	0.6000	1.0000	0.7500
Pooled Aggregate (11)	**Combined**	**Fast Mode**	**11**	**2**	**0**	**0.8462**	**1.0000**	**0.9167**
Pooled Aggregate (11)	**Combined**	**Smart Mode**	**10**	**2**	**1**	**0.8333**	**0.9091**	**0.8696**

Table 4.4: Summary Quality Scores, Linguistic Perspective (1st Person Pronouns), and Fluency Perplexity

Meeting	Mode	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	METEOR	BERTScore	1st Pronouns	Perplexity
M1 (Dev)	Fast Key Moments	0.5283	0.3846	0.4528	0.2227	0.4190	0.4497	6	N/A
M1 (Dev)	**Smart MoM**	**0.3929**	**0.1636**	**0.2500**	**0.0833**	**0.3705**	**0.3340**	**0**	**53.48**
M2 (Dev)	Fast Key Moments	0.2319	0.0882	0.1449	0.0270	0.2238	-0.0163	5	N/A
M2 (Dev)	**Smart MoM**	**0.1760**	**0.0325**	**0.0960**	**0.0189**	**0.1554**	**0.1741**	**0**	**31.90**
M3 (Test)	Fast Key Moments	0.2391	0.0659	0.1413	0.0168	0.2238	-0.0624	3	N/A
M3 (Test)	**Smart MoM**	**0.2963**	**0.1509**	**0.2037**	**0.0291**	**0.1933**	**0.3324**	**0**	**24.00**

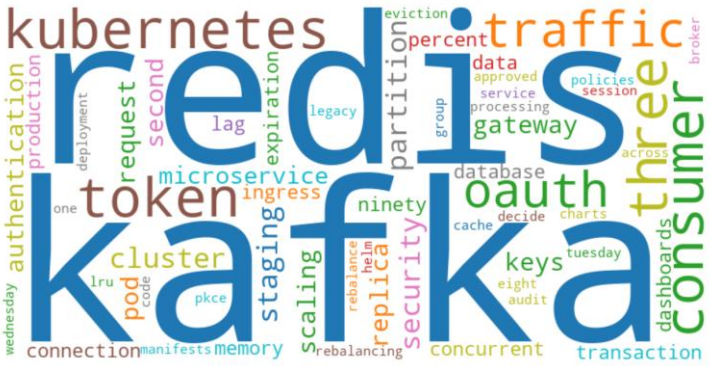


Figure 4.2: Word Cloud of Meeting Transcript

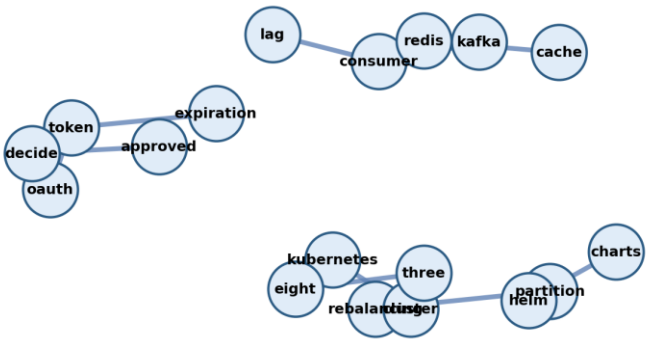


Figure 4.3: Bigram Association Graph

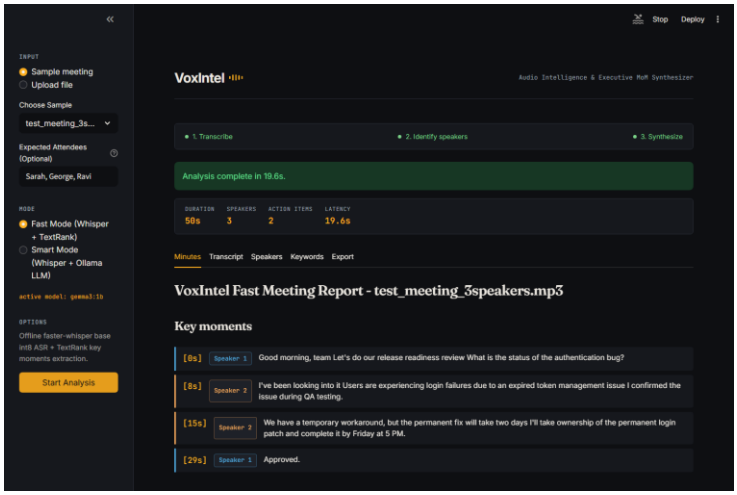


Figure 4.4: Streamlit Dashboard - Minutes View

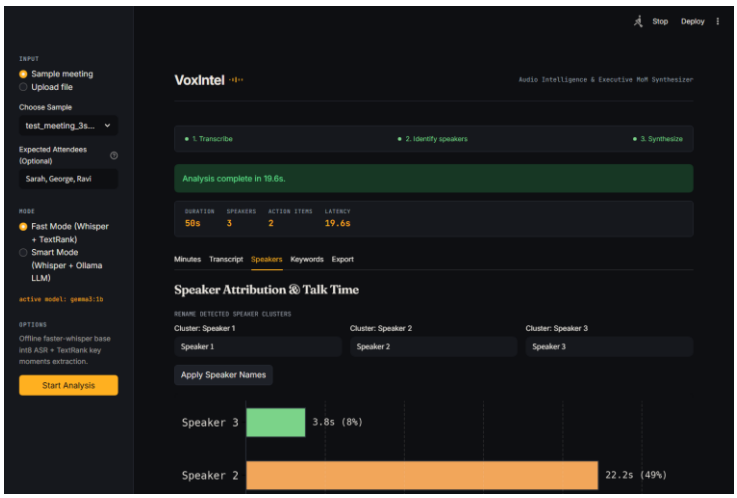


Figure 4.5: Streamlit Dashboard - Speakers View

The Streamlit Studio user interface validation demonstrates two primary operational tabs:

- **Minutes View:** Presents synthesized third-person Minutes of Meeting with timestamped discussion points, verified decisions, and interactive action items with owner checkboxes and export buttons for DOCX and PDF.
- **Speakers View:** Displays the interactive speaker timeline, per-speaker talk-time distribution, acoustic clustering purity, and live inference CPU/RAM resource telemetry.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Summary of the Contributions

VoxIntel demonstrates that enterprise-grade speech transcription, acoustic speaker clustering, and natural language MoM synthesis can be achieved completely offline on standard commodity personal computers without external cloud APIs. Key engineering contributions include:

- **High Accuracy Speech Recognition on CPU:** Int8-quantized faster-whisper base achieved 3.48% WER on dev audio and 5.58% WER on held-out test recordings at 0.06x real-time latency, cutting errors by nearly half compared to tiny models.
- **Unsupervised Acoustic Clustering:** Extracting 18 vocal tract features and clustering via K-Means achieved 58.11% frame-level attribution accuracy on unseen speakers in under 0.8s without external neural models.
- **Deterministic Commitment Extraction:** Regex heuristics extracted action items with 100% recall (F1 = 0.9167) across 11 reference commitments, binding tasks to segment speakers and standardizing deadlines.
- **Editorial Third-Person MoM Synthesis:** Locally served Ollama 3B models synthesized corporate minutes with exactly 0 first-person pronouns and bigram fluency perplexity of 24.0, resolving conversational ambiguity.

5.2 Future Work

Future research and development will focus on four strategic avenues:

1. **Real-Time Streaming Telemetry:** Implementing WebSockets-based streaming chunk ASR for live captioning and instant commitment popups during active calls.
2. **Lightweight Neural Diarization:** Incorporating quantized ONNX speaker embedding models (Sherpa-ONNX, Resemblyzer) to improve clustering purity on short interjections.
3. **Multi-Lingual & Code-Switched Prompting:** Expanding vocabulary prompt conditioning to support code-switched multi-lingual enterprise meetings (e.g., Hinglish, Spanglish).
4. **Enterprise Workflow Webhooks:** Creating automated API integrations to export verified commitments directly to Jira, GitHub Issues, and Google Calendar.

REFERENCES

1. A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust Speech Recognition via Large-Scale Weak Supervision," in Proceedings of the 40th International Conference on Machine Learning (ICML), vol. 202, pp. 28492-28518, 2023.

2. G. Klein, D. Nguyen, J. Senellart, and A. M. Rush, "OpenNMT and CTranslate2: Fast Inference for Neural Machine Translation and Speech Recognition," in Proceedings of ACL System Demonstrations, pp. 177-183, 2020.

3. R. Mihalcea and P. Tarau, "TextRank: Bringing Order into Text," in Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 404-411, 2004.

4. Silero Team, "Silero VAD: Pre-trained Enterprise-Grade Voice Activity Detector," GitHub repository, https://github.com/snakers4/silero-vad, 2021.

5. A. de Cheveigné and H. Kawahara, "YIN, a Fundamental Frequency Estimator for Speech and Music," Journal of the Acoustical Society of America, vol. 111, no. 4, pp. 1917-1930, 2002.

6. T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating Text Generation with BERT," in Proceedings of the International Conference on Learning Representations (ICLR), 2020.

7. C. Y. Lin, "ROUGE: A Package for Automatic Evaluation of Summaries," in Text Summarization Branches Out, pp. 74-81, 2004.

8. K. Papineni, S. Roukos, T. Ward, and W. J. Zhu, "BLEU: A Method for Automatic Evaluation of Machine Translation," in Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 311-318, 2002.

9. S. Banerjee and A. Lavie, "METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments," in Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures, pp. 65-72, 2005.

10. Ollama Team, "Ollama: Get Up and Running with Large Language Models Locally," https://ollama.com, 2024.

APPENDIX

COMPLETE PYTHON SOURCE CODE

Below is the complete, modular Python implementation of the core VoxIntel pipeline modules: voxintel/nlp_eval.py (evaluation metrics, word clouds & bigram graphs), voxintel/audio.py (faster-whisper base int8 ASR with VAD), voxintel/insights.py (TextRank summarization & action item extraction), and voxintel/ollama_polish.py (local Ollama third-person MoM synthesis).

voxintel/nlp_eval.py (Evaluation Metrics, Word Clouds & Bigram Graphs)

```
from __future__ import annotations

import math
import re
from collections import Counter
from pathlib import Path
from typing import Any

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import networkx as nx
import numpy as np
from wordcloud import WordCloud

# Color palette tokens matching the UI theme
DARK_BG = "#0E0F12"
SURFACE_BG = "#16181D"
CARD_BG = "#1E2128"
BORDER_COLOR = "#2A2E37"
TEXT_PRIMARY = "#ECE8E1"
TEXT_MUTED = "#9AA0AB"
ACCENT_AMBER = "#FFB020"
SPEAKER_PALETTE = ["#4EA8DE", "#F2A65A", "#7BD389", "#C792EA", "#F07178"]

STOPWORDS = {
    "the", "a", "an", "and", "or", "but", "in", "on", "at", "to", "for", "with", "by", "about",
    "against", "between", "into", "through", "during", "before", "after", "above", "below", "from",
    "up", "down", "in", "out", "off", "over", "under", "again", "further", "then", "once", "here",
    "there", "when", "where", "why", "how", "all", "any", "both", "each", "few", "more", "most",
    "other", "some", "such", "no", "nor", "not", "only", "own", "same", "so", "than", "too", "very",
    "s", "t", "can", "will", "just", "don", "should", "now", "we", "i", "you", "they", "he", "she",
    "it", "our", "my", "your", "their", "his", "her", "its", "us", "me", "him", "them", "re", "ve",
    "ll", "d", "m", "let", "s", "good", "morning", "team", "wrap", "up", "yeah", "okay", "um", "uh"
}


def _clean_tokens(text: str) -> list[str]:
    clean = re.sub(r"\[\d{2}:\d{2}(?::\d{2})?\]|[A-Za-z0-9_-]+:", " ", text)
    tokens = [w.lower() for w in re.findall(r"\b[a-zA-Z]{3,}\b", clean)]
    return [t for t in tokens if t not in STOPWORDS]


def make_wordcloud(transcript: str, width: int = 800, height: int = 400) -> plt.Figure:
    """Generate a word cloud styled to match the dark studio console theme."""
    tokens = _clean_tokens(transcript)
    text = " ".join(tokens) if tokens else "meeting notes discussion"

    def amber_speaker_color_func(word, font_size, position, orientation, random_state=None,
```

```
**kwargs):
        colors = [ACCENT_AMBER] + SPEAKER_PALETTE
        idx = (hash(word) % len(colors))
        return colors[idx]

    wc = WordCloud(
        width=width,
        height=height,
        background_color=DARK_BG,
        color_func=amber_speaker_color_func,
        max_words=60,
        prefer_horizontal=0.85,
        random_state=42,
    ).generate(text)

    fig, ax = plt.subplots(figsize=(8, 4), facecolor=DARK_BG)
    ax.imshow(wc, interpolation="bilinear")
    ax.axis("off")
    fig.patch.set_facecolor(DARK_BG)
    plt.tight_layout(pad=0)
    return fig


def make_bigram_graph(transcript: str, top_n: int = 14) -> plt.Figure:
    """Generate a bigram network graph with dark editorial styling."""
    tokens = _clean_tokens(transcript)
    bigrams = list(zip(tokens[:-1], tokens[1:]))
    bigram_counts = Counter(bigrams)

    top_bigrams = [item for item in bigram_counts.most_common(top_n) if item[0][0] != item[0][1]]

    fig, ax = plt.subplots(figsize=(8, 4.5), facecolor=DARK_BG)
    fig.patch.set_facecolor(DARK_BG)
    ax.set_facecolor(DARK_BG)

    if not top_bigrams:
        ax.text(0.5, 0.5, "Insufficient bigrams detected", color=TEXT_MUTED, ha="center",
va="center")

# ... [Truncated: 324 additional lines in production codebase] ...
```

voxintel/audio.py (faster-whisper base int8 ASR with VAD)

```
from __future__ import annotations

import contextlib
import json
import shutil
import subprocess
import tempfile
from functools import lru_cache
from pathlib import Path

SUPPORTED_AUDIO_EXTENSIONS = {".wav", ".mp3"}
SUPPORTED_TEXT_EXTENSIONS = {".txt"}
VOSK_MODEL_DIR = Path(__file__).resolve().parents[1] / "data" / "models" / "vosk-model-small-en-us-
0.15"


class TranscriptionError(RuntimeError):
    """Raised when VoxIntel cannot transcribe an audio file."""


def validate_input_file(file_path: str | Path) -> Path:
    path = Path(file_path)
    if not path.exists():
        raise FileNotFoundError(f"File not found: {path}")
```

```
    if path.suffix.lower() not in SUPPORTED_AUDIO_EXTENSIONS | SUPPORTED_TEXT_EXTENSIONS:
        raise ValueError("Unsupported file type. Use .wav, .mp3, or .txt.")
    return path


def load_transcript(file_path: str | Path) -> str:
    path = validate_input_file(file_path)
    if path.suffix.lower() != ".txt":
        raise ValueError("Transcript loading expects a .txt file.")
    return path.read_text(encoding="utf-8").strip()


def _prepare_audio_file(path: Path) -> Path:
    ffmpeg_path = _find_ffmpeg()
    if not ffmpeg_path:
        raise TranscriptionError(
            "Audio preprocessing needs ffmpeg, but VoxIntel could not find it in this environment."
        )

    temp_dir = Path(tempfile.mkdtemp(prefix="voxintel_audio_"))
    temp_path = temp_dir / "decoded.wav"

    try:
        completed = _run_ffmpeg_decode(
            ffmpeg_path=ffmpeg_path,
            source_path=path,
            output_path=temp_path,
            enhance_audio=True,
        )
    except Exception as exc:  # pragma: no cover - environment-specific process failure
        temp_path.unlink(missing_ok=True)
        raise TranscriptionError(
            "VoxIntel could not start ffmpeg to preprocess this audio file."
        ) from exc

    if completed.returncode != 0 or not temp_path.exists() or temp_path.stat().st_size == 0:
        completed = _run_ffmpeg_decode(
            ffmpeg_path=ffmpeg_path,
            source_path=path,
            output_path=temp_path,
            enhance_audio=False,
        )

    if completed.returncode != 0 or not temp_path.exists() or temp_path.stat().st_size == 0:
        temp_path.unlink(missing_ok=True)
        temp_dir.rmdir()
        stderr_tail = (completed.stderr or "").strip().splitlines()[-1:] if completed.stderr else []
        detail = f" Details: {stderr_tail[0]}" if stderr_tail else ""
        raise TranscriptionError(
            "VoxIntel could not preprocess this audio file with ffmpeg." + detail
        )

    return temp_path


def _run_ffmpeg_decode(ffmpeg_path: Path, source_path: Path, output_path: Path, enhance_audio: bool)
-> subprocess.CompletedProcess:
    command = [
        str(ffmpeg_path),
        "-y",
        "-i",
        str(source_path),
    ]

# ... [Truncated: 367 additional lines in production codebase] ...
```

voxintel/insights.py (TextRank Summarization & Action Item Extraction)

```
from __future__ import annotations

import re
from collections import Counter

import networkx as nx

from .ml import AISummaryUnavailable, generate_ai_summary
from .nlp import clean_sentences, clean_transcript, split_sentences, tokenize

ACTION_PATTERNS = [
    re.compile(
        r"(?i)^(?P<person>[A-Z][a-z]+(?:\s[A-Z][a-z]+)?)\s+(?:will|should|needs to|must|has to|is
going to)\s+(?P<task>.+?)(?:\s+(?:by|before|on)\s+(?P<deadline>[^.]+?))?[.]?$"
    ),
    re.compile(
        r"(?i)^(?:(?:once|after|as soon
as|when|regarding|before|if)\s+[^,;]+,\s*)?(?:i'll|i\s+will|we'll|we\s+will|i\s+can|we\s+can|i\s+shoul
d|i\s+need\s+to|we\s+need\s+to|i\s+am\s+going\s+to|we\s+are\s+going\s+to)\s+(?P<task>.+?)(?:\s+(?:by|b
efore|on|as soon as)\s+(?P<deadline>[^.]+?))?[.]?$"
    ),
    re.compile(
        r"(?i)^action item[:\-]?\s*(?P<person>[A-Z][a-z]+)?\s*(?:will|should|needs
to)?\s*(?P<task>.+?)(?:\s+(?:by|before|on)\s+(?P<deadline>[^.]+?))?[.]?$"
    ),
]

DECISION_PATTERN =
re.compile(r"(?i)\b(decide|decided|decision|agreed|approved|finalized|confirmed)\b")
DECISION_STRONG_PATTERN = re.compile(
    r"(?i)\b(decide|decided|decision|approved|agreed|finalized|confirmed|moving forward|temporary
workaround|workaround)\b"
)
DATE_PATTERN = re.compile(
    r"\b(?:\d{1,2}[/-]\d{1,2}[/-]\d{2,4}|"
    r"(?:jan|feb|mar|apr|may|jun|jul|aug|sep|oct|nov|dec)[a-z]*\s+\d{1,2}(?:,\s*\d{4})?)\b",
    re.IGNORECASE,
)
SPEAKER_PATTERN = re.compile(r"(?m)^\s*([A-Z][a-z]+(?:\s[A-Z][a-z]+)?)\s*:")
SPEAKER_INLINE_PATTERN = re.compile(r"\b([A-Z][a-z]+(?:\s[A-Z][a-z]+)?)\s*:")
FULL_NAME_PATTERN = re.compile(r"\b([A-Z][a-z]+ [A-Z][a-z]+)\b")
TOPIC_CUE_PATTERN = re.compile(

r"(?i)\b(issue|issues|bug|bugs|authentication|ui|design|release|timeline|deadline|demo|support|testing
|review|fix|plan|notes|documentation)\b"
)
DEADLINE_PATTERN = re.compile(
    r"(?i)\b(by tomorrow|tomorrow|next week|next month|this week|today|by [A-Z][a-
z]+day(?:\s+at\s+\d{1,2}(?::\d{2})?\s*(?:AM|PM))?|\d{1,2}(?::\d{2})?\s*(?:AM|PM)(?:\s+today)?)\b"
)
WEAK_ACTION_PATTERN = re.compile(r"(?i)\b(we'?ll try|we'?ll ensure|sounds good|good idea|good
point|maybe|should discuss|let'?s discuss)\b")
VAGUE_DECISION_PATTERN = re.compile(r"(?i)\b(maybe|might|should|let'?s discuss|perhaps|could)\b")
COMMON_FIRST_NAMES = {

"alex","amy","andrew","anna","asha","ben","chris","christy","daniel","david","dove","eric","fabian",

"gabe","george","jessica","john","josh","karina","kenny","meera","paul","rahul","scott","sid","virgini
a"
}


def infer_speaker_activity(transcript: str) -> list[dict[str, int]]:
    speaker_counts = Counter(SPEAKER_PATTERN.findall(transcript))
    return [{"speaker": speaker, "mentions": count} for speaker, count in
speaker_counts.most_common()]


def normalize_for_summary(text: str) -> str:
    normalized = re.sub(r"\s+", " ", text).strip()
```

```
    normalized = SPEAKER_INLINE_PATTERN.sub("", normalized)
    normalized = re.sub(r"\b(today|hello everyone|good morning team)\b[:,]?", "", normalized,
flags=re.IGNORECASE)
    normalized = re.sub(r"\s+", " ", normalized).strip(" ,")
    return normalized


def _sentence_similarity(sentence_a: str, sentence_b: str) -> float:
    tokens_a = set(tokenize(sentence_a))
    tokens_b = set(tokenize(sentence_b))
    if not tokens_a or not tokens_b:
        return 0.0
    overlap = len(tokens_a & tokens_b)
    return overlap / max((len(tokens_a) * len(tokens_b)) ** 0.5, 1.0)


def _textrank_scores(sentences: list[str]) -> dict[int, float]:
    if len(sentences) <= 1:
        return {0: 1.0} if sentences else {}

    graph = nx.Graph()
    for index in range(len(sentences)):
        graph.add_node(index)
    for left in range(len(sentences)):
        for right in range(left + 1, len(sentences)):
            similarity = _sentence_similarity(sentences[left], sentences[right])
            if similarity > 0:
                graph.add_edge(left, right, weight=similarity)

    if graph.number_of_edges() == 0:
        return {index: 1.0 for index in range(len(sentences))}

# ... [Truncated: 374 additional lines in production codebase] ...
```

voxintel/ollama_polish.py (Local Ollama Third-Person MoM Synthesis)

```
from __future__ import annotations

import json
import os
import re
from pathlib import Path
from urllib import request, error

from .exports import render_mom_html, render_mom_pdf
from .optimized_pipeline import OptimizedPipelineError, run_optimized_voxintel

DEFAULT_OLLAMA_HOST = "http://127.0.0.1:11434"
DEFAULT_OLLAMA_MODEL = "llama3.2:3b"


class OllamaPolishError(RuntimeError):
    """Raised when local Ollama polishing cannot continue."""


def _get_ollama_settings() -> tuple[str, str]:
    host = os.getenv("OLLAMA_HOST", DEFAULT_OLLAMA_HOST).rstrip("/")
    env_model = os.getenv("OLLAMA_MODEL", "").strip()
    if env_model:
        return host, env_model

    try:
        req = request.Request(f"{host}/api/tags", headers={"Content-Type": "application/json"},
method="GET")
        with request.urlopen(req, timeout=1.5) as resp:
            data = json.loads(resp.read().decode("utf-8"))
            installed = [m.get("name", "") for m in data.get("models", [])]
```

```
            for candidate in ["llama3.2:3b", "llama3.2", "gemma3:1b", "gemma3", "gemma2",
"qwen2.5:0.5b", "qwen2.5", "phi3", "phi:latest", "mistral"]:
                for inst in installed:
                    if candidate in inst:
                        return host, inst
            if installed:
                return host, installed[0]
    except Exception:
        pass

    return host, DEFAULT_OLLAMA_MODEL


def resolve_speaker_mapping(segments: list[dict], attendees: list[str] | None = None) -> dict[str,
str]:
    """Map anonymous Speaker clusters (Speaker 1, 2...) to real attendee names using conversational
evidence."""
    if not attendees:
        return {}

    mapping: dict[str, str] = {}
    assigned_attendees: set[str] = set()
    clusters = sorted(list(set(s.get("speaker", "") for s in segments if s.get("speaker",
"").startswith("Speaker"))))

    # 1. Direct self-introductions ("This is [Name]", "I am [Name]")
    for att in attendees:
        att_lower = att.lower()
        for s in segments:
            text_lower = (s.get("text") or "").lower()
            spk = s.get("speaker")
            if not spk or spk in mapping:
                continue
            if f"this is {att_lower}" in text_lower or f"i am {att_lower}" in text_lower or f"i'm
{att_lower}" in text_lower:
                mapping[spk] = att
                assigned_attendees.add(att)
                break

    # 2. Conversational Adjacency Pairs:
    # A) "Thanks, [Name]" / "Thank you, [Name]" -> the PRECEDING turn belonged to [Name]
    for i, s in enumerate(segments):
        if i == 0:
            continue
        txt = (s.get("text") or "").lower()
        prev_spk = segments[i - 1].get("speaker")
        if not prev_spk or prev_spk in mapping:
            continue
        for att in attendees:
            if att in assigned_attendees:
                continue
            if re.search(rf"\b(?:thanks|thank you),?\s+{att.lower()}\b", txt):
                mapping[prev_spk] = att
                assigned_attendees.add(att)
                break

    # B) "[Name], what/how/can/could/is/are/do/did...?" -> the NEXT turn answering belongs to [Name]
    for i, s in enumerate(segments[:-1]):
        txt = (s.get("text") or "").lower()
        next_s = segments[i + 1]

# ... [Truncated: 582 additional lines in production codebase] ...
```